

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Classwork 2
  - Due **tomorrow, 9/13 @5 pm**
- Homework 1
  - Available on Brightspace/GitHub Classroom
  - Due Friday 10/4 @5 pm

This Lecture

- Independence
  - Joint Independence Factoring
  - Structuring Independence
- Bayesian Networks
  - Structures
  - Semantics

Last Lecture

- Probability Theory

Propositional Logic	Probabilistic Logic
$KB \models \alpha$	$P(Q\|e)$
Truth Table -> Models	Truth Table -> Distribution

- Axiomatic Properties
  - Probability Value Range: $0 \leq P(w) \leq 1$
  - Sum of all probability values: $\sum_w P(w) = 1$
  - Law of Total Probability: $\mathrm{P}(\alpha) = \sum_{w \in M(\alpha)} P(w)$

Last Lecture

- Conditional Probability Distribution $P(Q|e)$
- Conditioning Rule (or Bayes' Theorem)

$$P(Q|e) = \frac{P(Q,e)}{P(e)} \text{ OR } P(Q,e) = P(Q|e)P(e)$$

World	L	T	$P(L,T)$	$P(T\|L)$
w_0	F	F	0.3	0.3/(0.3+0.1)=0.75
w_1	F	T	0.1	0.1/(0.3+0.1)=0.25
w_2	T	F	0.1	0.1/(0.1+0.5)=1/6
w_3	T	T	0.5	0.5/(0.1+0.5)=5/6

=1

Uncertainty

Scalability?

Scalability

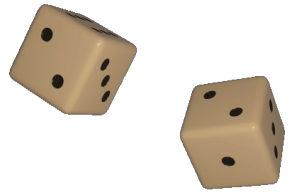
World	L	T	$P(L,T)$	$P(T\|L)$
w_0	F	F	0.3	0.75
w_1	F	T	0.1	0.25
w_2	T	F	0.1	1/6
w_3	T	T	0.5	5/6

World	L	T	R	$P(L,T,R)$
w_0	F	F	F	0.12
w_1	F	T	F	0.04
w_2	T	F	F	0.04
w_3	T	T	F	0.20
w_4	F	F	T	0.18
w_5	F	T	T	0.06
w_6	T	F	T	0.06
w_7	T	T	T	0.30

Does Law of Total Probability still apply?

Yes! $P(L,T) = \sum_{r \in R} P(L,T,r)$

Example



- Suppose we roll two, fair, 6-sided die: A, B.
  - Will the outcome of A tell us anything about the outcome of B? Does treating A as evidence update our beliefs about B?
    - No! E.g., A=6 does not affect P(B|A=6)
  - Then, what can be said about P(B|A)?
    - P(B|A) = P(B)

Independence

- Two variables A, B are said to be independent, written $A \perp\!\!\!\perp B$, if and only if the following holds:

$$A \perp\!\!\!\perp B \Leftrightarrow P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B)\,, \forall\, a \in A, b \in B$$

- Example: for two independent, fair, 6-sided dice A and B
  - How many rows in the distribution tables for P(A) and P(B)?
  - How many rows in their JPT (i.e., P(A,B))?
  - How does independence help with scalability?

Joint Independence Factoring

- For two independent variables A, B, their joint distribution probability can be **factored**:

$$A \perp\!\!\!\perp B \iff P(A, B) = P(A)P(B)$$

- Independence also allows us to **ignore irrelevant evidence**:

$$A \perp\!\!\!\perp B \iff P(A|B) = P(A)$$

- And help with **factoring large joint probability**
  - E.g., 36-row JPT -> two 6-row independent probability distribution

Conditional Dependence

- However, two independent variables A, B can be made **conditionally dependent or independent** when given evidence of another set of variables C.
- Example: two, fair, 6-sided die: A, B
  - Conditionally dependent
    - C=A+B
  - Conditionally independent
    - C=|A|

Structuring Independence

- **Goal**: find a **procedure** for breaking joint probability distribution into **factors** while navigating complex relationships between variables of independence
- Why?
  - Joint distribution **grows exponentially** as # of variables increases
  - Use the information of the joint **without** having to construct it or consult it explicitly
- How?

Bayesian Network

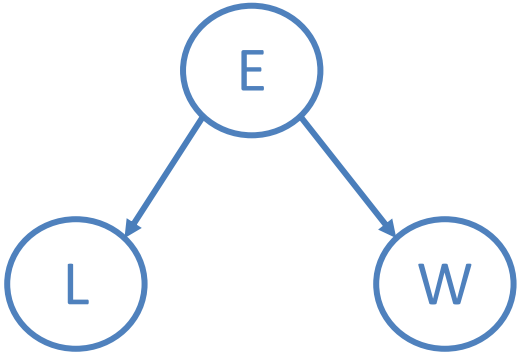
- **Bayesian Network** is a **data structure** used in probabilistic reasoning to encode independent relationships between variables in the reasoning system, and is composed of:
  - **Structure:** a Directed Acyclic Graph (DAG) modeling [cause] -> [effect] relationships between variables
    - **Nodes:** variables
    - **Edges:** point from causes (parents) to their direct effects (children)
  - **Probabilistic Semantics**: conditional probability distributions associated with each node

Bayesian Networks - Structure

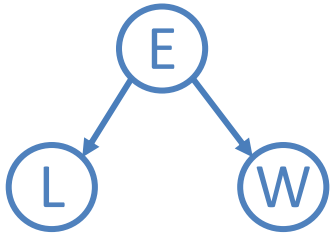
- Why **D**irected **A**cyclic **G**raph ?
  - **D**irected:
    - Edges are directional but not necessarily causal
    - An edge $X \rightarrow Y$ simply means that $X \not\perp\!\!\!\perp Y$
  - **A**cyclic:
    - To avoid time travel in DAG
    - Causes cannot lead to their own causes

Example

- Earthquake
  - E: an earthquake was detected
  - L: call from LMU
  - W: wireless emergency alert
- Using a DAG, how should we model the relationships of direct causes and effects between these three variables in our graph?



Example



Suppose $L \leftarrow E \rightarrow W$,

- Would we expect **causes** to be independent of their effects?
  - $E \perp\!\!\!\perp L$? and $E \perp\!\!\!\perp W$? No!
- Would we expect **effects of a cause** to be independent?
  - $L \perp\!\!\!\perp W$? Depends on whether we know the state of the cause.
- Would we expect **effects of a cause** to be independent once **we know the state of the cause**?
  - $L \perp\!\!\!\perp W | E$? Yes!

Model Sources

- Where do Bayesian Networks come from?
  - **Top-down** [Model -> Data]
    - Start with a programmer-curated **model** and then **interpret** witnessed **data** through the model.
  - **Bottom-up** [Data -> Model]:
    - Start with **data** and attempt to **learn** the **model** (i.e., structure relationships between variables) based on witnessed independence relationships.

Bayesian Networks - Semantics

- The semantics of a Bayesian Network define the rules by which probabilistic queries can be made from the network.
- Conditioning Rule:

$$P(Q|e) = \frac{P(Q,e)}{P(e)}$$

$$P(Q,e) = P(Q|e)P(e)$$

{% of Q and e} = {% of Q in worlds where e is true}{% of a world where e true}

Conditioning

- If $Q \perp\!\!\!\perp e$ (intuitively: the evidence is irrelevant to the query), then to what is $P(Q \mid e)$ equivalent?
  - $Q \perp\!\!\!\perp e \Leftrightarrow P(Q \mid e) = P(Q)$
- If $Q \perp\!\!\!\perp e_1 \mid e_2$ for two pieces of evidence e_1, e_2, then to what is $Q \perp\!\!\!\perp e_1 \mid e_2$ equivalent?
  - $Q \perp\!\!\!\perp e_1 \mid e_2 \Leftrightarrow P(Q \mid e_1, e_2) = P(Q \mid e_2)$

Conditional Independence

- Two variables α,β are said to be **conditionally independent** given some third set of variables γ (written $\alpha \perp\!\!\!\perp \beta \mid \gamma$) if and only if the following holds:

$$\alpha \perp\!\!\!\perp \beta \mid \gamma \Leftrightarrow P(\alpha \mid \beta,\gamma)=P(\alpha \mid \gamma)$$

- "Knowing about β tells me nothing about α that γ didn't already tell me"

Practice

Suppose that a system with 3 binary variables, X,Y,Z has the property that X⫫Z | Y. Showing each step, solve for **P(X=0|Y=0,Z=0)** and **P(X=0|Y=0)** separately.

Hint: You may need to use both conditioning and law of total probability.

$P(X,Y,Z)$	$X=0$		$X=1$	
	$Y=0$	$Y=1$	$Y=0$	$Y=1$
$Z=0$	0.10	0.05	0.15	0.20
$Z=1$	0.10	0.05	0.15	0.20

Bonus


Cumulative Distribution Function [CDF]
1.00
0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Violent Crime Index
Ice Cream Sales